

2D SCANNING LIDAR · LR-1BS SERIES

M12-CONNECTOR OLEI LR-1BS2-V2 270° LIDAR

The LR-1BS2 zone scanner on an M12 Ethernet connector. Same 270° sweep, same 16 banks of three levels, same four-in / four-out switching — with the connector most industrial platforms are already wired for.



■ WHY THIS SENSOR

- **M12 Ethernet connector** — 4-pin M12×1 A-code female, 2 m LAN — the industrial standard, not an M8 adapter chain
- **270° field of view from one corner mount** — a single sensor covers the front and both flanks of an AMR or a machine envelope
- **16 banks × 3 protection levels** — warn, slow and stop zones drawn per bank and switched live from the platform
- **4 switching inputs / 4 switching outputs** — select the active bank and drive the stop signal directly, with or without a PLC
- **Point cloud *and* switching output together** — azimuth, distance and signal strength over Ethernet while the relay logic keeps running
- **0.225° angle resolution at 10–25 Hz** — 16K–40K points per second, 130 ms response at 10 Hz
- **≤ ±1° horizontal plane error** — the scan plane stays where the drawing says it does
- **5 W at 12–24 V DC, IP65, 450 g** — platform power, wash-down tolerant, no fan

0.2–10 m

WORKING RANGE

270°

FIELD OF VIEW

16 × 3

ZONES / LEVELS

M12

ETHERNET CONNECTOR

■ APPLICATION SCENARIOS



AMR & AGV

270° obstacle avoidance from a single chassis-mounted scanner



INTRALOGISTICS

Forklifts, tuggers and conveyor zones in active aisles



INDUSTRIAL CELLS

Machine-envelope monitoring and presence sensing



OEM INTEGRATION

Volume platforms where size, weight and power are budgeted

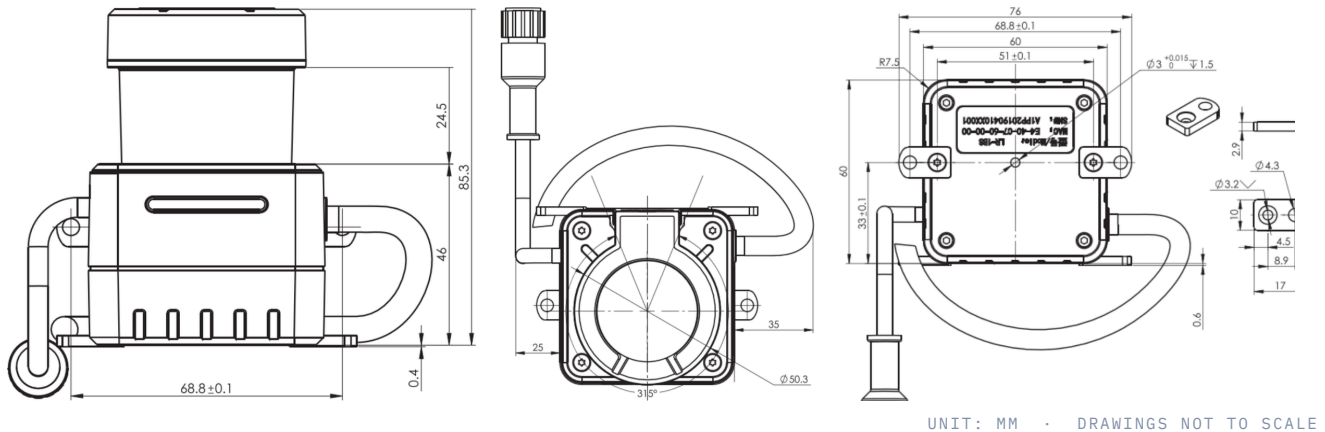
LR-1BS2-V2 — SPECIFICATIONS



PARAMETER	TECHNICAL DATA
Sensing technology	2D scanning ToF LiDAR
Laser wavelength	905 ± 20 nm
Laser safety level	Class 1 (IEC 60825-1:2014)
Laser channels	1
Field of view (horizontal)	270°
Beam divergence	5.8 mrad longitudinal 0.58 mrad transversal
Angle resolution (horizontal)	0.225°
Scanning frequency	10–25 Hz
Working distance (based on reflectivity)	0.2–10 m @ 80% reflectivity 0.2–8 m @ 10% · 0.2–2 m @ 4% · 0.2–10 m @ reflector
Accuracy ①	Absolute < ±50 mm Repeat < 20 mm
Distance resolution	1 mm
Signal strength	0–60000
Scanning rate	16K–40K pts/s
Horizontal error (horizontal angle)	±1°
Response time	130 ms @ 10 Hz

PARAMETER	TECHNICAL DATA
Number of bank fields	16 banks, 3 levels/bank
Switching inputs	4
Switching outputs	4
Ethernet connection	100 Mbps Ethernet
Connection	12-core cable, 2 m 4-pin M12×1 A-code female, 2 m LAN
Operating voltage	12–24 V DC
Power consumption @ 25 °C	5 W
Mechanical dimensions	≤ 60 × 60 × 85 mm
Weight	450 g
Degree of protection	IP65 or optional
Operating temperature	–10 to +50 °C
Storage temperature	–30 to +70 °C
Ambient light limit	80 000 lux
Indicator status	RGB × 4
Mounting	Bottom / back
Function software	Oforge
Basic software	OLAMViewer · ROS driver

■ OUTLINE AND MOUNTING



Certification. FCC, EMC, CE, RoHS, REACH, FDA. EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. ① Standard value; actual value depends on environmental conditions. Accuracy and working distance both degrade with lower target reflectivity — size the zones from the 4% and 10% figures, not the 80% figure. The V2 differs from the LR-1BS2 in its Ethernet connector (M12 rather than M8); the sensing specification is the same.

Installation. Mount the scanner so the 270° plane is unobstructed through its full sweep and the blind 90° faces structure, not traffic. Keep the optical window clear of spray and dust, clean it on a schedule in dirty environments, and verify every configured bank on the platform after any change to mounting height or angle.

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